

11

Unsteady Flow in Pipelines

Throughout the last 10 chapters we have addressed steady flow of liquids in pipelines. This means that the pipeline flow rates, velocities, and pressures at any point along the pipeline do not change with time. In other words, flow is not time-dependent. In reality this may not be true. Many factors affect pipeline operation. Even though volumes may be taken out of storage tanks at a constant rate and pump stations pump the liquid at a uniform rate there will always be some environmental or mechanical factors that could result in changes to the steady-state flow in the pipeline. In this chapter we briefly discuss unsteady flow and pipeline transients. A detailed analysis of unsteady flow and pipeline transient pressures is beyond the scope of this book. The reader is advised to refer to more specialized texts on unsteady flow, listed in the References section.

11.1 Steady Versus Unsteady Flow

In steady-state flow we assume the hydraulic gradient, representing the variation in pipeline pressures along the pipeline, to be fixed as long as the same liquid is being pumped at the same flow rate. Also, in steady-state flow we assume that liquids are incompressible. In reality, liquids *are* compressible, to some extent. The compressibility effect can cause

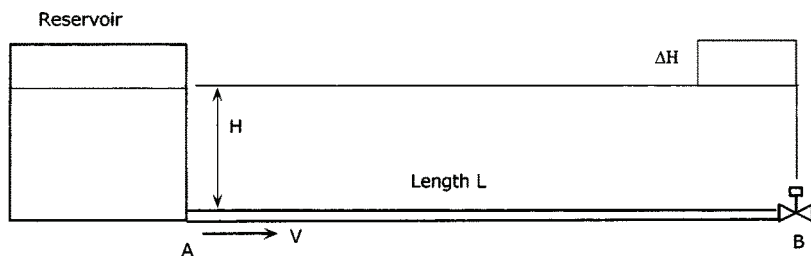


Figure 11.1 Transient with sudden valve closure.

transients or surges in otherwise steady-state flow. If, during the course of operation, different liquids enter the pipeline at different times, such as in a batched pipeline, the pressures, velocities, and flow rates will vary with time, resulting in changes to the hydraulic pressure gradient.

Unsteady flow occurs when the flow rate, velocity, and pressure at a point in a pipeline change with time. Examples of such pressure transients include the effect of opening and closing valves, starting and stopping a pump, or starting and stopping pipeline injections or deliveries. A basic example of unsteady flow would be a situation when a downstream valve on a pipeline is suddenly closed, as shown in Figure 11.1.

11.2 Transient Flow due to Valve Closure

To illustrate the effect of unsteady flow and transient pressure, we will consider a simple pipeline system from a storage tank at A, to a valve at the end of the pipe at B, as shown in Figure 11.1. The valve is located at a distance L from the tank. The pipeline is assumed to be horizontal, with no elevation changes and uniform diameter D . In order to simplify the problem, we will further assume that the friction in the pipe is negligible. Later we will consider friction and its effect on transient flow. Since friction is ignored, the steady-state hydraulic gradient line is horizontal.

Let us assume that initially, at time $t=0$, the valve at B is fully open, steady-state flow Q exists from A to B, and the liquid velocity throughout the pipe is V . The tank head is assumed to be H . The velocity is assumed to be positive in the downstream direction from A to B. If nothing changed and the valve at B were left open long enough, the tank head H would be dissipated by flow through the valve.

When we suddenly close the valve at B, we introduce a transient or surge into the pipeline. As the valve closes, pressure, velocity, and flow

variations occur both upstream and downstream of the valve. Since we are only interested in the pipe section from A to B, we will ignore the effect downstream of the valve at B. Closing the valve suddenly causes the velocity of flow at B to become zero instantaneously. This sudden drop in velocity at the valve causes the pressure head at B to increase suddenly by some value ΔH . This pressure rise is equal to the change in momentum of the liquid from a velocity of V to zero.

If the velocity changes from V to zero,

$$\Delta V = 0 - V = -V$$

The negative sign indicates a decrease in velocity. The magnitude of ΔH is given by

$$\Delta H = aV/g \quad (11.1)$$

where

a = Velocity of propagation of pressure wave, ft/s

V = Velocity of liquid flow, ft/s

g = Acceleration due to gravity, ft/s²

If instead of closing the valve fully, we close it such that the velocity drops from V to V_1

$$\Delta V = V_1 - V$$

The corresponding head rise at the valve is

$$\Delta H = a(V_1 - V)/g \quad (11.2)$$

The pressure head increase ΔH at B causes a slight expansion of the pipe and also an increase in liquid density, which depends upon the bulk modulus of the liquid, pipe size, and pipe material. For most pipelines, this pipe expansion and increase in liquid density (decrease in liquid volume) will be insignificant. Depending on the properties of the liquid and pipe, the pressure increase propagates upstream from the valve at the speed of sound (wave speed), a , generally in the range of 2000 to 5000 ft/s. The wave traveling at the speed of sound reaches the tank in time L/a . At this point in time the velocity of liquid throughout the pipe has become zero and, correspondingly, the pressure everywhere has reached $H + \Delta H$. The consequence of this is that the liquid has compressed somewhat and the pipe has stretched as well.

The above scenario is an unstable condition since the tank head is H but the pipeline head is higher ($H + \Delta H$). Due to this pressure differential, the liquid starts to flow from the pipeline into the tank with velocity $-V$.

The velocity is thus changed from zero to $-V$ which causes the pressure to drop from $(H+\Delta H)$ to H . Therefore, a negative pressure wave travels toward the valve B. Since friction is negligible, the magnitude of this reverse flow velocity is exactly equal to the original velocity V .

At time $t=2L/a$, the pressure in the pipeline returns to the steady-state value H , but reverse flow continues. However, since the valve is closed, no flow can take place upstream of the valve. Consequently, the pressure head drops by the amount ΔH , forcing the reverse flow velocity to zero. This causes the pipe to shrink and the liquid expands.

At time $t=3L/a$, this transient has reached the tank and the flow velocity is now zero throughout the pipe. The pipe pressure, however, has dropped to a value ΔH below that of the tank. This causes flow from A to B equal to the original steady-state flow. During this process the pipe pressure returns to the original steady-state value.

Finally, at time $t=4L/a$, the pressure wave from A has reached the valve at B and the flow velocity reaches the original steady-state value V . The total time elapsed ($4L/a$) is defined as one wave cycle, or the theoretical period of the pipeline. As the valve is completely closed, the preceding sequence of events starts again at $t=4L/a$. Since we assumed a frictionless system, the process continues indefinitely and the conditions are repeated at an interval of $4L/a$. However, in reality, due to pipe friction the transient pressure waves are dissipated over a definite period of time and the liquid becomes stationary with no flow anywhere and a pressure head equal to that of the tank.

Figure 11.2 shows a plot of the pressures at the valve at various times, beginning with the valve closure at time $t=0$ to time $t=12L/a$ (three wave periods). When friction losses are taken into consideration, the variation of pressure at the valve with time will be as shown in Figure 11.3. It can be seen from Figure 11.3 that, due to pipe friction, the transient pressure waves will die down over a definite period of time and the liquid will reach equilibrium with zero flow throughout the pipeline with the pressure head exactly equal to the initial tank head.

As mentioned before, examples of unsteady flow include filling a pipeline, power failure and pump shut-down, and opening and closing of valves. In summary we can say that unsteady flow conditions occur when the pipeline flow rate continually varies with time or variations in flow between two steady-state conditions occur. Unsteady flow conditions may be referred to as transient flow. Any change from a regular steady-state flow condition can cause transients. Slow transients are called surges. Depending on the magnitude of the transient and the rate at which the transient occurs, pipeline pressures may exceed steady-state pressures and sometimes violate the maximum allowable operating pressure (MAOP) in

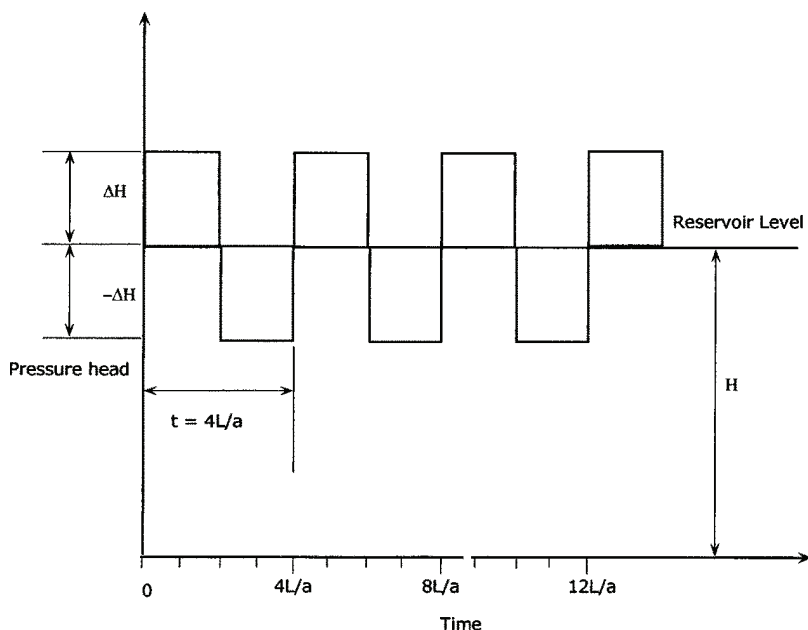


Figure 11.2 Pressure variation at valve: friction neglected.

a pipeline. Design codes and federal regulations dictate that the pressure surge cannot exceed 110% of the MAOP. Therefore, for a pipeline with an MAOP of 1400 psi, during transient situations the highest pressure reached anywhere in the pipeline cannot exceed $1400 + 140 = 1540$ psi.

To control surge pressures and avoid pipeline overpressure, relief valves may be used that are set to relieve the pipe pressure by allowing a certain amount of flow to a relief tank.

11.3 Wave Speed in Pipeline

We have seen from Equation (11.1) that the instantaneous head rise due to sudden stoppage of flow is

$$\Delta H = aV/g$$

The above equation can be derived easily considering a control element of flow in the tank-pipe example discussed in Section 11.2. However, we will not attempt to derive the equation here. The reader is referred instead to one of the fine books on unsteady flow listed in the References section. At

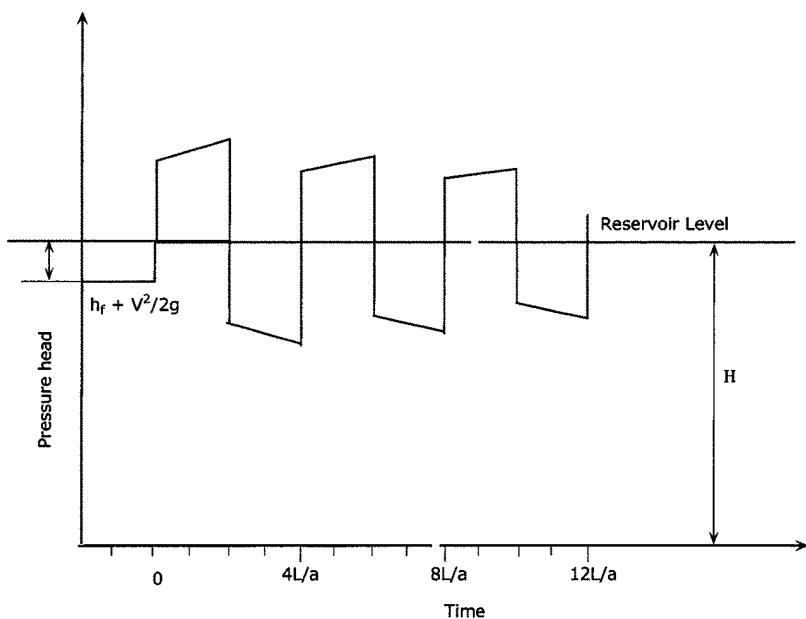


Figure 11.3 Pressure variation at valve: friction considered.

this point we will assume that the above equation is correct and proceed to apply it to some practical situations.

Since the transient head rise ΔH is directly dependent on the wave speed a from Equation (11.1), we must accurately determine its value for each pipeline.

It has been found that the wave speed in a liquid depends on the density and bulk modulus of the liquid. It depends also on the elasticity of the pipe material, pipe diameter, and wall thickness. Also, if the liquid contains free air or gas, the wave speed will be affected by these. An approximate value of the wave speed is calculated from

$$a = (K/\rho)^{1/2}$$

where

a = Wave speed, ft/s

K = Bulk modulus of liquid, psi

ρ = Density of liquid, slugs/ft³

However, the above value of wave speed does not take into account the elastic nature of the pipe. Hence in the following discussion the

equation will be modified to account for the pipe material and its elastic properties.

Consider an extreme case of a pipeline system where the pipe is considered rigid and the liquid incompressible. Incompressible liquid means infinite wave speed, which is impossible. Since no pipe is totally rigid, the infinite wave speed situation is not possible.

According to Wylie and Streeter (see [References](#)), a general equation for calculating the wave speed in terms of liquid properties, pipe size, and elasticity of pipe material is as follows:

$$a = \frac{(K/\rho)^{1/2}}{[1 + C(K/E)(D/t)]^{1/2}} \quad (11.3)$$

where

a=Wave speed, ft/s

K=Bulk modulus of liquid, psi

ρ =Density of liquid, slugs/ft³

C=Restraint factor, dimensionless

D=Pipe outside diameter, in.

t=Pipe wall thickness, in.

E=Young's modulus of pipe material, psi

An equivalent formula in SI units is

$$a = \frac{(K/\rho)^{1/2}}{[1 + C(K/E)(D/t)]^{1/2}} \quad (11.4)$$

where

a=Wave speed, m/s

K=Bulk modulus of liquid, kPa

ρ =Density of liquid, kg/m³

C=Restraint factor, dimensionless

D=Pipe outside diameter, mm

t=Pipe wall thickness, mm

E=Young's modulus of pipe material, kPa

The Young's modulus is also referred to as the modulus of elasticity of the pipe material.

The restraint factor C depends on the type of pipe condition. The following three cases are possible:

Case 1: Pipe is anchored at the upstream end only

Case 2: Pipe is anchored against any axial movements

Case 3: Each pipe section is anchored with expansion joints

The calculated value of the wave speed depends on the restraint factor C which in turn depends on the type of pipe support described above. It is found that the restraint type has less than 10% effect on the magnitude of wave speed.

The restraint factor C is defined in terms of the Poisson's ratio μ of the pipe material as shown below (see [Wylie and Streeter](#) in References):

$$C=1-0.5\mu \text{ for case 1} \quad (11.5)$$

$$C=1-\mu^2 \text{ for case 2} \quad (11.6)$$

$$C=1.0 \text{ for case 3} \quad (11.7)$$

where μ =Poisson's ratio for pipe material, usually in the range of 0.20 to 0.45 (for steel pipe $\mu=0.30$).

It must be noted that the above values of the restraint factor C are applicable to thin-walled elastic pipes with a pipe diameter to thickness ratio (D/t) greater than 25. For thick-walled pipes with a D/t ratio less than 25, the following C values may be used (see [Wylie and Streeter](#) in References):

Case 1

$$C = \frac{2t}{D}(1 + \mu) + \frac{D(1 - 0.5\mu)}{D + t} \quad (11.8)$$

Case 2

$$C = \frac{2t}{D}(1 + \mu) + \frac{D(1 - \mu^2)}{D + t} \quad (11.9)$$

Case 3

$$C = \frac{2t}{D}(1 + \mu) + \frac{D}{D + t} \quad (11.10)$$

Example Problem 11.1

Water flows through a steel pipe at velocity of 6 ft/s. A valve at the end of the pipe is partially closed and the velocity instantly reduces to 4 ft/s. Estimate the surge pressure rise at the valve. Use a wave speed of 3000 ft/s for the water in the pipe.

Solution

From Equation (11.1):

$$\begin{aligned}\text{Pressure rise} &= aV/g \\ &= 3000 \times (6-4)/32.2 = 186.33 \text{ ft}\end{aligned}$$

Example Problem 11.2

A steel pipe carries water at 20° C. The pipe diameter is 600 mm and wall thickness 6mm. Assume Young's modulus $E=207$ GPa and Poisson's ratio $=0.3$. Use 1000 kg/m^3 for the density of water and a bulk modulus $K=2.2$ GPa. Calculate the wave speed through this pipeline considering the three pipe constraints.

Solution

The D/t ratio is

$$600/6=100$$

therefore thin-walled pipe values for restraint factor C apply. Considering the pipe to be completely rigid, the wave speed is

$$a=(K/\rho)^{1/2}=(2.2 \times 10^9/1000)^{1/2}=1483 \text{ m/s}$$

(a) For a pipe anchored at the upstream end only, from Equation (11.5):

$$C=1-0.5\mu=1-0.5 \times 0.3$$

or

$$C=0.85$$

Therefore,

$$\begin{aligned}a &= \frac{(K/\rho)^{1/2}}{(1 + CKD/Et)^{1/2}} \\ &= \frac{1483}{[1 + 0.85 \times 2.2 \times 600/(207 \times 6)]^{1/2}} \\ &= \frac{1483}{1.9034} = 1075 \text{ m/s}\end{aligned}$$

(b) For a pipe anchored against any axial movement, from Equation (11.6):

$$C = 1 - \mu^2 = 1 - (0.3)^2 = 0.91$$

$$a = \frac{1483}{[1 + 0.91 \times 2.2 \times 600 / (207 \times 6)]^{1/2}}$$

$$= 1057 \text{ m/s}$$

(c) For a pipe with each section anchored with expansion joints, from Equation (11.7):

$$C=1$$

and

$$a = \frac{1483}{[1 + 1.0 \times 2.2 \times 600 / (207 \times 6)]^{1/2}}$$

$$= 1033 \text{ m/s}$$

11.4 Transients in Cross-Country Pipelines

Cross-country pipelines and other trunk pipelines transporting crude oil, refined products, etc., are several hundred miles long and generally have multiple pump stations located along the pipeline. Average pump station spacing may be 30 to 60 miles or more. The pump stations are designed to provide the necessary pressure to overcome friction in the pipeline between pump stations and any static head lift necessary due to pipeline elevation differences.

The study of transient pressures in long-distance pipelines, called surge analysis, is fairly complex due to the fact that the frictional pressure drops in the pipelines are large compared with the instantaneous pressure surge due to sudden stoppage of flow.

A rigorous mathematical analysis of the pressure versus flow variations due to transient conditions results in partial differential equations that need to be solved with specified boundary conditions. Such solutions are difficult, if not impossible, using manual methods, even with a scientific calculator. However, over the years engineers and scientists have simplified these equations using finite difference methods and other mathematical approximations to result in the solution of simple simultaneous equations at every nodal point along the pipeline. Of course, the more the pipeline is subdivided, the higher is the accuracy of

calculation. Today, with high-powered desktop computers, we can model these transient pressures in pipelines using the Method of Characteristics, which has proven to be quite accurate when compared with field tests. Several commercial computer programs are available to model transient pressures in pipelines due to valve closure, pump start-up and shut-down, and other upset conditions.

The potential surge is the instantaneous pressure rise due to sudden stoppage of a flow, such as closing of a valve at the end of the pipeline. This pressure was shown from Equation (11.1) to be

$$\Delta H = aV/g$$

The increase in storage capacity of a pipeline due to increase in pressure is termed line packing. Line packing occurs when the flow in a pipeline is changed by gradual or sudden closure of a valve. Consider a long crude oil pipeline flowing at a steady flow rate of Q bbl/hr. Due to sudden closure of a valve downstream there is a pressure rise immediately upstream of the valve. A pressure wave with an amplitude of ΔH travels from the valve upstream towards the pump station. Oil flows downstream of the wave front, pressure rises gradually at the downstream end, and the hydraulic grade line gradually reaches the horizontal level. During this time more oil gets packed between the valve and the pressure wave, hence the term “line pack.” The pressure rise at the valve is made up of two components:

1. The potential surge pressure, ΔH , from Equation (11.1) due to instantaneous closure of the valve
2. The pressure rise due to line pack, ΔP

The pressure rise ΔP due to line pack in a long oil pipeline may be several times the size of the potential surge ΔH , depending on the length of the pipeline, its diameter, etc.

Attenuation is the process by which the amplitude of the surge pressure wave is reduced as it propagates along the pipeline due to the reduction in the velocity differential across the wave front. This is because, even though the wave front has passed a certain location on the pipeline, the oil continues to flow toward the valve. This causes a reduction in the velocity differential across the wave front as the wave propagates upstream.

In addition, the amplitude of the surge is also decreased due to pipe friction. This reduction, however, is a smaller component than that caused by the velocity differential across the wave front.

For a detailed analysis of transient analysis in long pipelines, see [Wylie and Streeter](#) in the References section.

11.5 Summary

In this chapter we have introduced the concept of unsteady flow in pipelines. If the velocity, flow rate, and pressure at any point in a pipeline change with time, unsteady flow is said to occur. Closing and opening valves, starting and stopping pumps, and injecting volumes or changing products all contribute to some form of unsteady flow. Some unsteady flow situations can cause pressure surges or transients that, when superimposed on the existing steady-state pressure gradient, may cause overpressure of a pipeline. Design codes for petroleum pipelines dictate that such transient overpressures cannot exceed 110% of the maximum allowable operating pressure in a pipeline.

We illustrated the concept of unsteady flow using a simple example of a tank connected to a pipe with a valve at the end. Sudden closure of the valve causes a pressure rise at the valve and generates a pressure wave that travels in the opposite direction of the flow at the speed of sound in the liquid. It was shown that flow rate and pressure variations occur throughout the pipeline due to the pressure wave traveling back and forth until it is attenuated by friction and equilibrium is established after a sufficient period of time. The wave speed depends mainly on the liquid bulk modulus and density. It also depends upon the pipe size and the elastic modulus of the pipe material.

Transient pressures developed in long-distance pipelines are generally modeled using computer programs based on the method of characteristics. This approach reduces the complex partial differential equations to simpler simultaneous equations which are solved sequentially from node to node, by subdividing the pipeline into equal segments. For more detailed analysis of unsteady flow the reader is referred to one of the publications listed in the References section.

11.6 Problems

- 11.6.1 Crude oil flows through a 16 in. diameter, 0.250 in. wall thickness steel pipeline, at a flow rate of 150,000 bbl/day. A valve located at the end of the pipeline is suddenly closed. Determine the potential surge generated and the wave speed in the pipeline assuming the pipe is anchored throughout against axial movement. Bulk modulus of crude oil=280,000 psi, specific gravity=0.89. Modulus of elasticity of steel= 30×10^6 psi. Poisson's ratio=0.3.
- 11.6.2 Compare the wave speed in a rubber pipeline 250 mm in diameter, 6 mm wall thickness carrying water, with a similar

pipeline 60 mm thick. $E=0.1$ GPa, $\mu=0.45$, $K=2.2$ GPa, $\rho=1000$ kg/m³. Assume the pipe is anchored at one end only.

- 11.6.3 A steel pipeline 12.75 in. in diameter and 0.250 in. wall thickness is used to transport water between two storage tanks. Calculate the wave speeds and restraint factor C using thin-walled and thick-walled formulas.